AM Session: EDA Theory

Kindly Refer attached google collab file for coding part

EDA is the process of "getting to know" your data before you jump into machine learning.

1. Basic Exploration & Cleaning
Why .head() and .tail()? These allow you to verify if the data loaded correctly and see the variety of entries at both ends of the file.

Handling Missing Values: In the Titanic dataset, the Age column often has missing entries. Filling them with the **median** is a standard practice because it prevents the loss of 500+ rows of data while keeping the statistical distribution stable.

2. Descriptive Statistics
The .describe() function: This is your primary tool for understanding the "spread" of your data.
Mean: The average value.
Std (Standard Deviation): How much the data varies. A high std in the Fare column means there is a large gap between cheap and expensive tickets.
Min/Max: The absolute range of your data.

3. Categorical Analysis

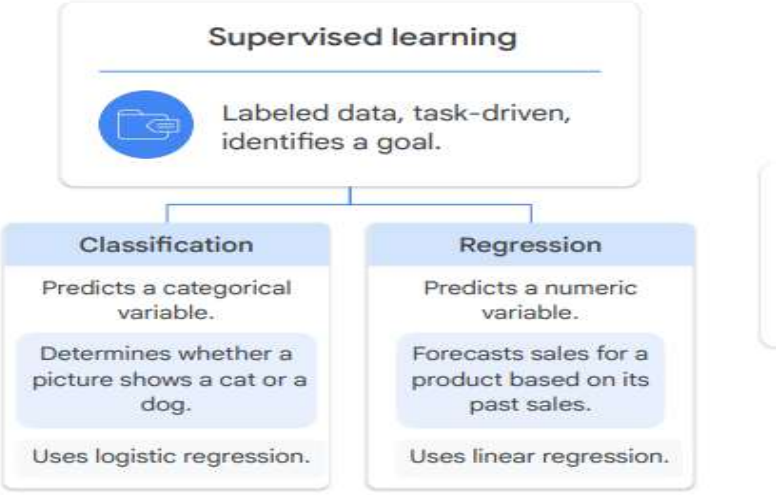
- **Unique Values & Frequency:** For columns like Survived or Sex, you need to know how many categories exist and if the data is "imbalanced" (e.g., if there are way more men than women).

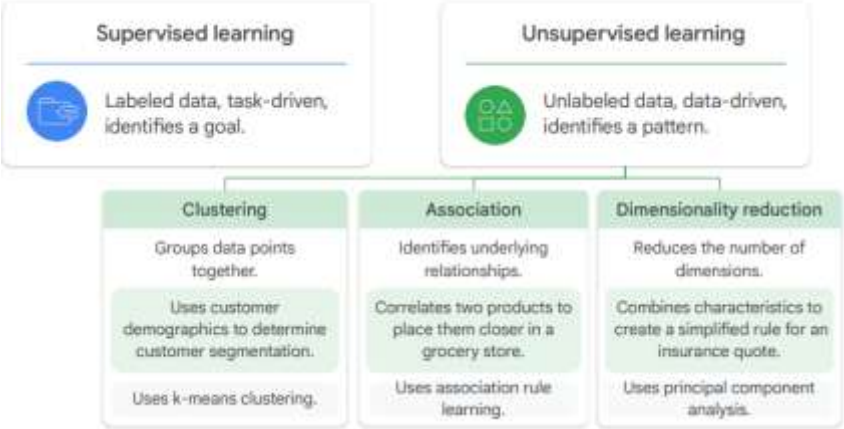
PM Session: Machine Learning Types

1. The Three Main Categories
Supervised Learning: The model learns from "labeled" data (where the answer is already known).
Example: Predicting if an email is **Spam** based on millions of emails already marked as spam.

Unsupervised Learning: The model finds hidden patterns in "unlabeled" data.
Example: Grouping customers by **purchasing behavior** to see who buys similar items.

Reinforcement Learning: An agent learns by interacting with an environment to get the highest **reward**.





Practical Implementation

Filtering: Using multiple conditions (like Age > 30 and Sex == 'female') helps you zoom in on specific demographics.
Feature Engineering: Creating a FamilySize column (by adding SibSp + Parch) provides a single, more powerful data point for the model than two separate columns.
Feature/Target Separation: In Supervised Learning, you must separate your features (**X**) from what you want to predict (**y**) so the model knows what to "study" and what the "exam answer" is.

Github link : https://github.com/sedoCdA/Day-25